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E-223 EMERGENCY DIESEL ENGINE-GENERATOR

1. SCOPE OF WORK

The works shall include the design, supply, manufactory testing, delivery to the site, installation, commissioning and testing of the Emergency Diesel Engine Generator in accordance with Part 4, Subsection 1.2 Summary of Work.

The Contractor shall be installed one (1) Diesel Generator and sized to supply continuously the essential load of 3 blocks (Block 2,3,4).

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the followings specific standards, or other equivalent International Standards.

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia
(General Regulation of Electrical Installation Indonesia).
- SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60034 : Rotating Electrical Machines

2.3 American Society of Mechanical Engineers

- ASTM A106 : Carbon Steel

2.4 American National Standards Institute and Institute of Electrical and Electronics Engineers

- ANSI/IEEE 12 : General Principles for Temperature Limits in the Ratings of Electric Equipment and for the Evaluation of Electrical Insulation.
- ANSI/IEEE 115 : Test Procedures for Synchronous Machines.
- ANSI C50.13/IEEE 126 : Rotating Electrical Machinery- Cylindrical Rotor Synchronous Generators.

2.5 National Fire Protection Association

NFPA 70	:	National Electrical Code
NFPA 30	:	Flammable and Combustible Liquids Code
NFPA 37	:	Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines

2.6 National Electrical Manufacturers Association

NEMA MG-1-22	:	Motors and Generators
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2.7 International Standards Organization

ISO 3046	:	Performance reciprocating internal combustion engines
ISO 8528	:	Reciprocating internal combustion engine driven alternating current generating

3. DESIGN AND OPERATING CONDITIONS

3.1 General

In case of a unit blackout a safety shutdown of the plant has to be ensured. The power provided from the batteries and emergency power generating system shall be used for safety shutdown of the power plant.

The essential loads for safety shutdown of the power plant consist of the following types: turning gear, essential oil pumps, battery chargers, standby/emergency lighting, security and fire alarms, and UPS, etc. The emergency diesel engine generator (EDG) set shall be capable of supplying continuously the essential loads.

The actual capacity of the EDG set shall be determined by the Contractor when the actual loads are fixed. The EDG set shall be designed for high reliability since it shall be provided to supply power under emergency conditions.

The EDG set shall comprise a water cooled industrial diesel engine, driving a flange mounted brushless revolving field generator, mounted together with a cooling radiator on a heavy duty fabricated channel base frame.

Control facilities shall be provided by a floor standing panel arranged to give automatic start-up on normal power supply failure and manual operation.

3.2 Method of Operation

Essential loads shall be connected to essential MCCs, as required. Where three (3) essential MCCs connected to one (1) common essential MCC, fed by a changeover arrangement such that when the normal power supply is lost, the essential MCC will be fed from EDG through common essential MCC.

When a loss of normal power supply signal is received by the EDG set, it shall automatically start. If the engine fails to start, two (2) further attempts shall be made, after which a “failure to start” alarm is raised. During the operation, the output and running conditions of the EDG set shall be continuously monitored.

The EDG set shall be designed for indoor installation and will consist of a diesel engine coupled to a generator. The EDG set shall be complete with all auxiliary equipment required to operate the EDG set locally as well as to monitor from the central control room.

3.3 Diesel Engine

The turbocharged diesel engine shall be designed for operation on HSD oil.

The following accessories shall be included for the EDG set:

- (1) Exhaust driven turbocharger, if required, to meet design conditions
- (2) Engine driven scavenging blower, if required, to meet design conditions
- (3) After coolers, exhaust silencer, exhaust fitting and flange, dry exhaust manifold/gas pipe, crankcase breather, and intake expansion joint
- (4) Heavy-duty engine mounted air intake filter-dry type, with service indicators
- (5) Flame arrester and silencer
- (6) Engine coolant water expansion tank
- (7) Engine mounted radiator with cooling fan(s) and radiator discharge duct flange
- (8) Expansion joints for all external connections
- (9) Engine coolant with rust inhibitor to match design conditions

- (10) Lubricating oil heat exchanger, if required.
- (11) Lubricating oil filters
- (12) Engine driven main lubricating oil, fuel oil, coolant water pumps and radiator cooling fans
- (13) Fuel pumps, HSD oil day tank with dike and fuel piping between day tank and DG set
- (14) DC motor driven start-up fuel priming and prelube oil pumps
- (15) Fuel filters with service indicators, with water in-fuel indication
- (16) Flexible fuel lines
- (17) Fuel cut-off valves
- (18) Engine governor
- (19) Foundation bolts and nuts

3.4 Ratings

The diesel engine-generator set load rating in terms of continuous rating and/or short-time overload rating shall be as follows:

- (1) Diesel engine-generator set shall be capable to supply 3 units of essential loads to the power plant for safety shutdown and additional loads i.e.:

- a. **Steam generating system essential loads**

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- b. HRSG essential loads
 - c. Air conditioning for computer rooms and control room
 - d. Communication systems and security systems
 - e. Other essential loads including, but not necessarily be limited to,
 - DC loads
 - Essential lighting

- (2) Diesel engine generator set shall be capable of starting the induction motor load while voltage and frequency are maintained within specified limits. Sizing calculations shall be supplied by the Contractor for approval by the Owner.

- (3) The continuous rating of the diesel engine-generator set shall be the electric

power output capability measured in kW at 0.8 power factor that the diesel engine-generator unit can maintain for 2,000 hours continuously without excessive vibration or engine damage.

- (4) The short-time overload rating of the diesel engine-generator set shall be the electric power output capability measured in kW at 0.8 power factor that the diesel engine-generator unit can maintain for two (2) hours continuously out of 24 consecutive hours without excessive vibration or engine damage.
- (5) The diesel engine generator set shall be capable in every respect to be operated in parallel with the power plant auxiliary system.
- (6) The diesel engine-generator shall be capable of meeting all ratings under the actual Site conditions, given in Part 4, Subsection 1.1 Description of the Project, Subsection 2.14 Seismic Requirements and Subsection 2.16 Environmental Consideration for the Site meteorological, seismic and environmental conditions, respectively.

3.5 Engine Emission and Sound Levels

Engine emission shall meet all Indonesian government regulations and the sound level requirements shall be as specified in Part 4, Subsection 2.9.

4. DETAILED REQUIREMENTS

4.1 Design and Construction

The diesel engine-generator set shall be designed to meet the following requirements:

- (1) The diesel engine-generator set will be used for standby emergency 400 V, three-phase, four-wired power and shall have the ability to start and achieve rated speed and voltage and be ready to pick up the load sequence specified in Clause 3 within ten (10) seconds of a start signal. After achieving rated speed and voltage, it shall be capable of maintaining voltage and frequency within specified limits under all loading conditions including short-time overload.
- (2) The diesel engine-generator set shall be complete with all accessory components as required by the Contractor's design for successful operation,

such as safety shutdown devices, control panel, lubricating oil system, fuel oil system, radiator or heat exchanger for engine jacket water system, automatic starting system, etc.

- (3) The generator shall be directly connected to the engine flywheel with a coupling and the flywheel casing shall be connected to the generator frame with an adapter to ensure constant alignment of the engine with generator. The generator shall be mounted on a common base with the engine. The flywheel and couplings shall be provided with suitable guards conforming to the applicable OSHA requirements.
- (4) The diesel engine-generator set shall carry its continuous rated load within the temperature rise at a maximum ambient temperature of 40°C.
- (5) The diesel engine-generator set shall be designed for reliable quick start capability at the ambient temperatures as Specified in Part 4, Subsection 1.1.
- (6) The diesel engine-generator set speed shall not exceed 1,000 rpm.
- (7) Linear vibration of the complete engine-generator set shall not exceed 5 mils, peak-to-peak under all loading conditions measured at the base of the set.
- (8) Vibration spring mounts or vibration isolator pads shall be furnished to limit structural vibration.
- (9) Insulation for the exhaust system silencer and piping shall be furnished and installed by the Contractor.
- (10) The Contractor shall furnish auxiliary power to the Contractor's control panel for supplying power to the battery charger, etc. All necessary transformers, circuit breakers, fuses, starters, wiring, etc., shall be provided by the Contractor to adapt this source to associated accessories.
- (11) A water sprinkler system shall be installed by the Contractor as fire protection in the room housing the engine-generator set. The engine-generator set shall be designed such that inadvertent operation of the sprinkler system will not cause damage to the diesel, cause the diesel to

become inoperative in any way, or adversely affect the availability or operability of the engine-generator set.

4.2 Diesel Engine Design

Engine drive shall be of the diesel type and shall conform to all applicable requirements set forth in this Specification.

The engine shall be of the four-stroke cycle multicylinder type and shall have a continuous rating at its governed speed of not less than the overload kW rating of the generator locked rotor load. The engine rating shall equal or exceed the generator motor starting capability.

Diesel engine shall be steel base mounted, pressure lubricated and shall include mufflers, filters, strainers, silencers, overspeed automatic shutdown equipment, alarms for high coolant temperature and low lube oil pressure and other necessary accessories as may be required, including battery racks and cables.

The engine shall maintain its speed at the overload rating of the engine-generator set.

The engine shall be a full diesel and shall be designed for satisfactory operation on HSD oil.

The diesel engine and accessories shall be designed to accommodate convenient maintenance.

Fuel System

The fuel system shall conform to NFPA Fire Codes 30 and 37.

The fuel oil system shall include an engine-shaft-driven fuel pump and a duplex-type fuel filter. The fuel pump shall be of the positive-displacement type. Piping for the suction of the fuel pump and return piping from the fuel pump and injectors shall be run to one (1) point on the side of engine. Flexible connections shall be provided for interconnection to the fuel oil piping.

The fuel oil day tank with adequate capacity of dike shall be provided by the Contractor to sustain a minimum of 12 hours of operation at highest possible capacity.

Cooling System

Engine cooling shall be by a closed cooling system using a radiator. The Contractor shall ensure. The radiator fan is adequately sized to overcome ductwork pressure drop.

Engine coolant shall be cooled in the heat exchanger shell and circulated by a shaft-driven coolant pump. Heat exchanger tube material shall be 90-10 Cu-Ni or equivalent. Engine block temperature shall be maintained by thermostat control.

Lubrication System

The engine lubrication system shall conform to NFPA 30 and 37.

The lubricating oil system shall include an engine shaft driven oil pump, a full flow oil strainer on the pump suction side and a full flow oil filter on the pump discharge side. All oil piping shall be pitched for gravity drainage.

All oil piping 50 mm OD and smaller shall be ASTM A 106 Grade B Sch. 80. All piping 80 mm OD and larger shall be ASTM A 106 Grade B Sch 40.

Cleaning of oil pipe shall be as specified in Technical Specification M-001 Piping.

Exhaust and Air Intake Systems

A residential type exhaust silencer shall be provided. The silencer shall be constructed of welded steel with metallic type acoustic material and shall resist to corrosion.

A dry replaceable cartridge type intake air filter and silencer shall be provided with the diesel engine. The diesel engine shall derive its combustion air directly from the room in which it is installed.

Starting System

The Contractor shall provide a dual electrical start system consisting of two (2) electrical starters and associated batteries for starting the engine, complete with battery stand, battery charger(s), charge control and automatic voltage regulator, ammeter, and thermal cutout switch in suitable IPW 55 enclosures with adequate provisions for ventilation.

Starting and control batteries (lead-acid type) with rack and accessories shall be provided for the diesel engine. Batteries shall be sized for starting the diesel engine and for furnishing all control power and shall be sufficient to operate the engine starter through three (3) complete series. Each starter and battery shall be capable of starting the diesel engine independently.

Battery charging equipment shall be of the current limiting, silicon diode, full wave rectifier, voltage-regulated type.

5. INSTRUMENTATION AND CONTROLS

5.1 Controls for Diesel Engine-Generator

The control panel shall be mounted in the Emergency Diesel-Generator Building. The DCS for Diesel Engine Generator shall be connected to Plant Bus for remote monitoring in the central room. The local control panel shall be designed and constructed to include the following features:

- (1) Start-stop controls
 - a. Automatic start on a signal from the transfer switches of the power plant auxiliary electrical system.
 - b. Manual start and stop shall be made possible by one-touch system on common control desk which located in central control room.
 - c. Manual start and shutdown from the local control panel.
- (2) AC output ammeter on the auxiliary mimic board of local control panel.
- (3) AC output wattmeter and watt-hour meter on the auxiliary mimic board of local control panel.
- (4) AC output voltmeter on the auxiliary mimic board of local control panel.
- (5) Frequency meter on the auxiliary mimic board of local control panel and an independent rpm meter on the auxiliary mimic board.
- (6) Running-time meter (total hour indication) on the auxiliary mimic board of local control panel
- (7) Manual synchronizing panel complete with two (2) volt meters, two (2)

frequency meters and sychronoscope.

- (8) Panel light with switch for emergency operation from the starter battery.
- (9) Local alarm indications shall be provided for automatic start, automatic shutdown, overcrank, engine overspeed, loss of control power, low diesel fuel level, low lube oil pressure on engine running, high cooling water temperature, generator overcurrent trips, and AC and DC failure.
- (10) Gauges for lube oil pressure and water coolant temperature.
- (11) Cranking sequence timer with controls set for single 30-second cranking cycle with adjustable timers to provide five (5) 10-second cranking cycles with 10 seconds between cycles. **(not applicable)**
- (12) The control panel shall have “start-stop-auto” controls for local manual start and stop and automatic start and stop. When the engine starts automatically, it shall continue to operate until manually shut down at the local control panel. Controls shall include an adjustable cooldown timer to allow no load operation before normal shutdown.

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A keylock switch shall be provided at the control panel to lock out remote controls and prevent starting while the engine is being serviced.

The engine generator shall be provided with a speed governor having adequate overspeed protection. The governor shall have maximum speed regulation of 2.5% through the load range of the generator. The speed governor shall be Woodward Hydraulic or equal approved by the Owner. The speed governor shall be equipped with reversible DC motor for speed adjustment from the control panel. The motor shall be suitable for use with the DC voltage available from the control storage battery. Overspeed device shall be included, and it shall be arranged to shut down the unit on excessive overspeed.

The Contractor shall provide radio noise suppression equipment, if required, to meet the limits prescribed by the local regulation or to prevent interference with the operation of any of the equipment.

Following an automatic start, the diesel engine-generator set shall be shut down by means of one (1) of the followings:

- (1) Local key lock switch.

- (2) Overspeed shutdown device.
- (3) Overcurrent trips.
- (4) Low oil pressure.
- (5) High temperature.

6. ELECTRICAL SPECIFICATION

6.1 Generator and Exciter Design

The generator shall be designed, constructed and tested in accordance with the latest applicable Standards of ISO, IEEE, ANSI and NEMA.

The generator shall be three-phase rated, with the adequate capacity *), 0.8 power factor 400 V, 4-wire, 50 Hz with the neutral brought out for connection. The generator neutral shall be solidly grounded. The generator shall be of single bearing type with flywheel pivoted generator shaft.

Note: *) The Contractor shall be responsible for final sizing of diesel engine-generator capacity to meet all essential load requirement and taking into account 25% spare capacity. The sizing calculation shall be submitted for review and approval by the Owner.

The diesel generator shall be designed to maintain the generator frequency at 50 Hz +1% to-2% for continuous operation at any load. The transient frequency variations shall be $\pm 5\%$ of 50 Hz.

The generator continuous rating shall be based on the temperature rise defined in ANSI C50. 13 for Class F with class H insulation corrected for an ambient temperature of 40°C.

6.2 Generator Protection

The control board shall be provided with the following generator protective relays:

- (1) Phase overcurrent (Voltage controlled)
- (2) Neutral grounding overcurrent and other protection as per standards.

6.3 Regular Test

The diesel generator shall be provided with a synchronizing facility for exercising the machine. This provision shall have facilities to synchronize and load the generator with the unit loads for regular test.

7. PERFORMANCE TESTING AND GUARANTEES

The completely assembled equipment shall be given a simulated operational test at the factory in order to guarantee its operation, including load rating and fuel consumption.

After the equipment has been put in service, the Contractor shall run a check test to ascertain that the equipment is performing properly in every respect.

8. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirements of Part 4, Section 2 subsection 2.7. The procedure specified shall include, but not be limited to, the followings:

- (1) Welding
- (2) Non destructive tests
- (3) Heat treating
- (4) Packaging and shipping
- (5) Shop test procedures
- (6) Cleaning and painting

The documentation specified shall include, but not be limited to the followings:

- (1) Shop Test Report and Site Test Report
- (2) Material Certification
- (3) NDE Report

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